Towards Reproducible Evaluation of Multistakeholder Recommender Systems: Data sets, Stakeholders, and Values

Cornelia M. Kloppers*
23958502@sun.ac.za
Stellenbosch University
Stellenbosch, South Africa

Jan H. van Vuuren
Stellenbosch University
Stellenbosch, South Africa
vuuren@sun.ac.za

ABSTRACT

As recommender systems increasingly serve multiple actors with diverse and often conflicting objectives, there is growing interest in multistakeholder evaluation of these systems. The integration of such perspectives into reproducible research remains limited due to the lack of structured guidance on how stakeholder values can be operationalised using public data sets. We address this gap by introducing a data-driven framework that links stakeholder values with concrete data requirements. Building on the conceptual foundations of the Dagstuhl Seminar 24211, we propose a data taxonomy for recommender systems and apply it to 23 publicly available data sets. Based on this analysis, it is possible to infer which stakeholder roles can be supported meaningfully, which values are evaluable via measurable offline proxies, and which data elements are required for principled evaluation. The methodology supports both data-driven and value-driven evaluation design and offers practical guidance for data set selection, proxy justification, and transparent reporting. We demonstrate its utility by highlighting representative data set examples. By bridging the gap between abstract value-oriented evaluation and real-world data constraints, this work supports more transparent, reproducible, and stakeholder-aligned evaluation practices in recommender systems.

CCS CONCEPTS

• **Information systems** → **Recommender systems**.

KEYWORDS

Recommender Systems, Multistakeholder Evaluation, Stakeholder Values, Offline Evaluation, Data sets, Reproducibility

ACM Reference Format:
Cornelia M. Kloppers and Jan H. van Vuuren. 2025. Towards Reproducible Evaluation of Multistakeholder Recommender Systems: Data sets, Stakeholders, and Values. In . ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 INTRODUCTION

Recommender systems are integral parts of modern digital platforms in which they guide users towards relevant and personalised content across diverse domains such as e-commerce, media streaming, and social networks [29, 40]. Their evaluation is a critical component of system development and deployment, thereby necessitating rigorous methodologies for ensuring their effectiveness and reliability [8].

Recommender system research has, traditionally, prioritised user-centric optimisation [4, 39]. Evaluations, however, often rely on algorithmic or system-level metrics, such as root mean square error (RMSE), normalised discounted cumulative gain (NDCG), or precision, as proxies for user satisfaction. While these metrics have been foundational in advancing the field, they are increasingly recognised as insufficient for capturing the broader socio-technical dynamics in which recommender systems function [41, 49].

In response, recent research has embraced a multistakeholder perspective by acknowledging that real-world recommender systems affect and serve a range of actors beyond end users [4, 39]. These include, but are not limited to, content providers, platform operators, and broader societal actors [12, 39]. Each stakeholder group may have distinct, and often conflicting, objectives [43, 49]. In an online marketplace, consumers may, for instance, seek relevant and reliable product suggestions, but the providers or sellers of those products may desire fair exposure of their items, and the platform operating the marketplace may aim to maximise overall engagement or revenue.

There is a critical and largely unmet need for evaluation protocols that are comprehensive in addressing these diverse stakeholder interests but also systematic and reproducible [8, 17]. The evaluation of *multistakeholder recommender systems* (MSRSs) is inherently complex: It requires the integration of multiple, potentially conflicting values, the operationalisation of these values into measurable metrics, and the incorporation of diverse data types corresponding to different stakeholder groups. These challenges further exacerbate existing reproducibility issues in recommender systems research, particularly in respect of the inconsistent application of evaluation practices and the lack of transparency in data set selection, preprocessing, and usage [9, 17, 49].

In this paper, we address the need for reproducible MSRS evaluation by introducing a structured, data-driven methodology that links stakeholder values with concrete data requirements. Building on the conceptual foundations established by the Dagstuhl Seminar 24211 [8], we propose a comprehensive data taxonomy for recommender systems and apply it to 23 publicly available data sets in order to identify their available data elements. By utilising the proposed data taxonomy, it is possible to determine (i) which stakeholder groups are meaningfully supported and (ii) which of their corresponding values can be evaluated in terms of measurable offline proxies, based on the availability of specific data elements.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
Conference'17, July 2017, Washington, DC, USA
© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-x-xxxx-xxxx-x/YY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

Our work bridges the gap between abstract value-based evaluation principles and the practical constraints of publicly available data sets. It supports both data-driven (bottom-up) and value-driven (top-down) evaluation design, equipping researchers and practitioners with a flexible framework for selecting data sets, justifying proxies, and documenting assumptions. Ultimately, this approach is aimed at improving the transparency, reproducibility, and stakeholder relevance in recommender system evaluations.
Our main contributions are as follows:

- We introduce a data taxonomy for recommender systems which consolidates diverse data types into a coherent schema.
- We analyse 23 widely used public data sets using this taxonomy and publish extended documentation on data elements via an open-access GitHub repository[1].
- We link stakeholder values to measurable offline evaluation proxies and derive the data requirements needed to support each value.
- We propose a practical framework for selecting data sets or designing evaluations based on either available data (bottom-up) or desired stakeholder values (top-down).

2 BACKGROUND

This section contains a concise review of existing work related to MSRSs.

2.1 Multistakeholder Recommender Systems

Rather than optimising recommender systems solely from the perspectives of the consumer, researchers are increasingly recognising that consumers are not the only parties affected by, or involved, in the recommendation process [4, 11, 39, 51]. In real-world settings, recommender systems are associated with stakeholders pursuing diverse and often conflicting objectives [5, 49]. On a video streaming platform, for instance, content creators might seek visibility for their videos, the platform might aim to maximise watch time, and users will expect recommendations that are aligned with their viewing preferences. This has led to the emergence of MSRSs, which aim to incorporate the needs and preferences of multiple stakeholder groups across the full lifecycle of system development [4].

The perspectives of stakeholder groups may be incorporated into the design, implementation, and evaluation of recommender systems. Multistakeholder design may be achieved by adopting participatory design approaches that actively involve external stakeholders in the design process [4]. Furthermore, the objectives of multiple parties may be considered in the recommendation algorithm itself, either by explicitly modelling and optimising for multiple stakeholder objectives or by re-ranking recommendations based on stakeholder preferences [4, 39]. Finally, the evaluation of recommender systems may be expanded by assessing the impact of recommendations on all relevant stakeholder groups. Multistakeholder evaluation involves identifying the relevant stakeholders, articulating their values and goals, and selecting appropriate evaluation metrics [11].

Abdollahpouri *et al.* [4] defines a recommendation stakeholder as "any group or individual that can affect, or is affected by, the delivery of recommendations to users." Across the literature, there is broad agreement on the primary stakeholder groups involved in recommender systems. Most notably, Jannach and Zanker [30], Abdollahpouri *et al.* [4], Zangerle *et al.* [49], and de Biasio *et al.* [19] consistently identified consumers, providers, and the system as the core stakeholders in the recommendation process. While early research largely focused on these groups, the report from a recent seminar on evaluation perspectives in recommender systems (Dagstuhl Seminar 24211 [8]) extended this taxonomy to encompass a broader and more nuanced set of roles. These include upstream, downstream, and third-party stakeholders. The following stakeholder group descriptions are based on the taxonomy proposed in the Dagstuhl Seminar [8] and include illustrative examples of their roles across application domains:

Consumers. The end-users who receive and act upon recommendations, such as customers browsing products on an e-commerce site or viewers selecting content on a streaming platform. They are the primary focus of traditional recommender system research, which often prioritises user satisfaction and engagement.

Providers. The individuals or entities that create or supply the recommended items. Depending on the domain, this may include authors, sellers, musicians, or course instructors.

System. The organisation or platform that operates the recommender system. This includes e-commerce companies, media services, or job-matching platforms, whose objectives often include revenue generation, user retention, or market growth.

Upstream stakeholders. Actors who contribute indirectly to the creation of recommended items or are affected on the provider side. For instance, in music recommendation, upstream stakeholders may include songwriters or producers who are entitled to royalties but are not directly featured in the recommendation itself.

Downstream stakeholders. Individuals or groups affected by the consumption of recommended items, despite not being the direct recipients. An example is children impacted by educational content recommended to their teachers.

Third-party stakeholders. External entities who have no direct interaction with the system, but whose interests or mandates intersect with its operation. These may include regulators, consumer protection agencies, advocacy groups, or other institutions concerned with compliance, ethical considerations, or broader societal outcomes.

The expanded taxonomy above facilitates a more holistic understanding of stakeholder dynamics in recommender systems and supports evaluation practices that better reflect the complex, socio-technical contexts in which these systems operate.

2.2 Stakeholder Values

Values are the general principles or norms that guide system behaviour by indicating what the system ought to promote or preserve, such as monetary reward, usefulness, or fairness. A stakeholder's values are shaped by its role, expectations, and contextual objectives within the system. These values give rise to stakeholder-specific

[1] available at https://anonymous.4open.science/r/Multi-Stakeholder-RecSys-Datasets-6FD5

goals, which are concrete, measurable aims derived from the interpretation of values in a given context. Monetary reward may, for example, translate into a goal of maximising revenue, while fairness might translate into ensuring balanced exposure across providers. These goals function as intermediate objectives and can be evaluated by invoking specific metrics such as, total revenue generated from recommendations, click-through rate as a proxy for usefulness, or exposure share as a measure of provider fairness.

Traditional methodologies which solely focus on user relevance or predictive accuracy, are increasingly being replaced or supplemented by approaches that embrace a broader range of values. These include, but are not limited to, research streams in:

Multi-objective recommender systems. These systems explicitly optimise more than one conflicting objective within a single stakeholder group, such as balancing relevance and diversity for users [39]. Some authors have recently extended this to optimise objectives across multiple stakeholders, effectively bridging into multistakeholder recommendation scenarios [28].

***Value-aware recommender systems* (VARS).** These systems extend traditional approaches by explicitly incorporating business objectives and accounting for multiple stakeholders, with the aim of optimising economic value [19]. A subset of VARS, economic recommender systems, focuses on organisational profitability by leveraging price and profit data [20]. This includes profit-aware systems which balance consumer relevance with provider profitability, and price-aware systems which model user price sensitivity to improve conversion and revenue [16, 18, 20, 52].

Fairness-aware recommender systems. These systems address the need to mitigate stakeholder inequalities, with focus areas including consumer-side, provider-side, and multi-sided fairness [35, 47]. Fairness is integral to multistakeholder recommendation and closely aligned with responsible artificial intelligence principles [5, 21].

Green recommender systems. These systems seek to minimise environmental impact by employing energy-efficient algorithms that reduce CO_2 emissions [10]. They may trade off accuracy for sustainability, aligning with the long-term objectives of external stakeholders such as environmental regulators.

There is clearly a wide range of research streams that implicitly reflect the values of different stakeholders. Due to inconsistent terminology and fragmented conceptualisations, however, it remains difficult to navigate or compare this evolving landscape systematically.

The report on the Dagstuhl Seminar [8] provides a comprehensive yet extendable taxonomy of values that offers a framework for articulating and evaluating the often implicit goals embedded within existing recommendation research. This taxonomy groups values into two overarching categories:

Economic and Business Values. These include, but are not limited to, profit, customer loyalty, market development and efficiency, which is typically emphasised by platform operators and providers.

Societal and Human-Centric Values. These may include fairness, well-being, trust, transparency, and environmetal sustainability.

In the context of recommender systems, most of these values are self-explanatory, and therefore we refer the reader to the Dagstuhl Seminar report for a more in-depth description of values [8].

Each stakeholder group may prioritise different combinations of these values. The problem is, however, that values are not inherently measurable. Their operationalisation into evaluation metrics requires specific types of data. For example, assessing fairness in exposure requires provider identifiers and impression logs. Measuring user satisfaction may require engagement data or subjective feedback.

2.3 Data Limitations in MSRSs

Although interest in MSRSs is growing, research progress is hindered by limited data availability. Methods for optimising multiple stakeholder objectives, such as techniques in multi-objective optimisation or fairness-aware modelling, require representative data about the stakeholders involved [4]. Likewise, evaluation depends on the availibility of stakeholder-specific features or identifiers. Most public data sets, however, lack such information and this poses a significant barrier to empirical research in this area [4, 12].

These data limitations have prompted researchers to rely on a range of ad-hoc workarounds. In the absence of data sets that contain multistakeholder attributes—such as provider metadata, price data, or sensitive demographic information—researchers often simulate stakeholder roles, augment data sets with proxy features, or repurpose data from other domains.

A common example is the widespread use of the MovieLens data set in multistakeholder research, despite its lack of provider and economic metadata [19]. Several authors have attempted to overcome this by randomly assigning movies to synthetic providers [43], or by imputing pricing and profit information using probabilistic methods [19]. Others have optimised for distributional fairness or product diversity by taking item popularity as a proxy for value, without modelling providers explicitly [21]. Researchers have also resorted to linking external public data sets, purchasing metadata, or relying on inferred attributes. These approaches, however, raise both ethical and technical concerns, including the risk of reinforcing bias and the misinterpretation of sensitive identity attributes [8]. While these techniques facilitate experimentation, they often rely on unverifiable assumptions that limit the generalisability and reproducibility of results.

Recent calls for improved data set documentation emphasise the need for greater transparency and standardisation in describing data provenance within recommender system research [39]. Inspired by practices such as datasheets for data sets [24], this includes detailed reporting of the origin, structure, and intended use of a data set to help researchers assess its suitability. Without such clarity, it remains difficult for researchers to assess whether a data set supports robust, stakeholder-aware evaluation or reinforces existing limitations in the field. These problems underscore the need for either richer public data sets or systematic frameworks for understanding and leveraging existing data in multistakeholder settings.

3 RECOMMENDER SYSTEM DATA TAXONOMY

Data sets form the foundation for both the development and evaluation of recommender systems. We examined a curated set of 23 data sets commonly employed in recommender system research in order to better understand the potential of publicly available data sets for multi-stakeholder evaluation. These were primarily selected from the list of data sets maintained by the ACM Recommender Systems (RecSys) Conference[2]. Additional consideration was afforded to domain diversity and relevance so as to ensure broad coverage across user, content, and platform types.

The data sets were manually reviewed to identify their available data fields. Given the observed variation in data set composition, it has become necessary to establish a structured classification of data elements.

To this end, we propose the taxonomy presented in Figure 1, which categorises the data elements commonly found in recommender system data sets. This taxonomy serves as a foundation for understanding the modelling and evaluation potential of each data set. Its application to the selected data sets is summarised in Table 1, which highlights the presence or absence of each data element and illustrates common gaps across the current landscape of public data sets.[3]

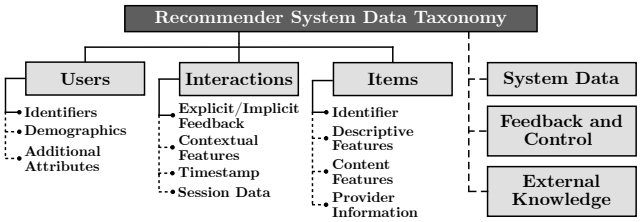

Figure 1: Taxonomy of data elements in recommender system data sets.

Present across all the data sets are the minimum data requirements for a recommender system: User identifiers, item identifiers, and a record of interactions between them. These data elements form the basis for modelling user preferences and generating recommendations, and is denoted by solid lines in Figure 1. While user-item interactions represent the primary data foundation, a recommender system may incorporate other types of data such as system-level information, feedback and control signals, and external knowledge—denoted by dashed lines in Figure 1.

Data on users, items and interaction may be enriched with additional information. For each of the primary elements—users, items, and interactions—additional data are represented as subcomponents, denoted by dotted lines in Figure 1.

Users. User data may be enriched by the inclusion of *demographic* information, such as age, gender, occupation, education level, and geographical location. The taxonomy further categorises any other data fields related to users, such as behavioural traits (*e.g.* browsing frequency), as *additional attributes*. This classification aligns with the recent work of Masciari et al. [34], who grouped user attributes into four broad categories: Personal, professional, behavioural, and environmental. Personal and professional attributes are presented as demographics, while behavioural and environmental attributes are presented as additional attributes. Among the data sets reviewed, 8 include demographic information and 7 provide additional user attributes.

Items. The proposed taxonomy classifies additional item-level data into three subcomponents: Descriptive features, content features, and provider information. *Descriptive features* refer to structured metadata that characterise the item, such as categories, tags, and genres. *Content features* include unstructured or semi-structured data—such as textual descriptions, images, audio, or video—from which item properties can be inferred by computational methods. *Provider information* captures any metadata about the creators, sellers, or other stakeholders associated with items (*e.g.*, author identifies or publisher names). All of the data sets reviewed include a data field in at least one of these three subcategories.

Interactions. Interaction data capture the observed engagement between users and items, and may be classified into four subcomponents: Feedback signals, contextual features, timestamps, and session data. *Feedback* signals encompass both explicit feedback—such as ratings, likes, or reviews—and implicit feedback, which includes behavioural signals such as clicks or purchases, serve as a proxy for user preference. *Contextual features* describe the situational aspects underlying the interaction (*e.g.*, device type or location). Although *timestamp* is a form of contextual information, it is treated as a distinct subcomponent due to its importance in enabling sequential and temporal analyses. Lastly, *session data* refer to groupings of interactions occurring within a single user session and may include a session identifier as well as session-level metadata (*e.g.*, session start time or duration). Among the data sets reviewed, 10 included explicit feedback, 15 capture implicit feedback, 15 contained contextual features, 19 include recorded timestamps, and 5 included session information or allow for session segmentation to be derived.

The three secondary elements in the proposed data taxonomy are less frequently available in publicly released data sets. This is largely because they are not essential for building a functional recommender system and often involve sensitive or proprietary information. Nonetheless, when available, these elements can provide valuable context for modelling and enable advanced evaluation strategies.

System data. This data element captures information about the internal functioning and configuration of the recommendation platform. Relevant data may include infrastructure and resource attributes (*e.g.*, latency, server load, energy usage), *catalog metadata* describing the state of the item pool (*e.g.*, stock availability or release windows), algorithm metadata related to the recommendation models deployed, and interface configuration data indicating how recommendations are presented (*e.g.*, whether a carousel or list layout is used).

[2]https://github.com/ACMRecSys/recsys-datasets
[3]An extended version of Table 1 is available at https://anonymous.4open.science/r/Multi-Stakeholder-RecSys-Datasets-6FD5.

Table 1: Binary presence of data attributes across data sets using taxonomy-based schema. An x indicates presence.

Data sets		Users		Items			Interactions					Secondary			
Name	**Domains**	**D**	**A**	**DF**	**PI**	**CF**	**EF**	**IF**	**T**	**SD**	**Cx**	**S**	**FC**	**SG**	**O**
MovieLens32M [26]	Media and Entertainment			×		×	×	×	×						×
MovieLens1M [26]	Media and Entertainment	×		×		×	×		×						
Netflix [36]	Media and Entertainment			×		×	×		×						
KuaiRec [23]	Media and Entertainment		×	×	×			×	×		×			×	
Steam [32]	Media and Entertainment		×	×	×	×	×		×		×				
Last.FM [13]	Media and Entertainment				×	×		×	×		×			×	
Spotify MPD [14]	Media and Entertainment	×	×	×	×	×		×			×				
KGRec [38]	Media and Entertainment			×		×		×							
MIND [48]	News and Information			×		×		×	×				×		×
Adressa [25]	News and Information	×	×	×	×	×		×	×	×*	×				
EB-NeRD [33]	News and Information	×	×	×		×		×	×	×	×				
Goodreads [45, 46]	Publishing and Literature			×	×	×	×	×	×		×			×	
Bookcrossing [42]	Publishing and Literature	×		×	×	×	×								
Endomondo [37]	Healthcare and Wellness	×		×				×	×		×				
EdNet-KT4 [15]	Education and E-learning			×				×	×	×*	×				
Amazon2023 [27]	E-commerce and Retail			×	×	×	×		×		×				
AmazonM2 [31]	E-commerce and Retail			×	×	×		×		×					
Epinions [22]	E-commerce and Retail			×			×		×		×			×	
RetailRocket [1]	E-commerce and Retail			×				×	×						
Diginetica [2]	E-commerce and Retail			×		×		×	×	×	×				
Synerise [44]	E-commerce and Retail			×		×		×	×		×				
Yelp [7]	Travel and Hospitality	×	×	×	×	×	×		×		×				
Tripadvisor [3]	Travel and Hospitality	×	×	×	×	×	×		×		×				
Total		8	7	22	11	18	10	15	19	5	15	0	1	4	2

Note
* session data may be derived from timestamps.
D: Demographics, **A:** Additional attributes, **DF:** Descriptive features, **PI:** Provider information, **CF:** Content features, **EF:** Explicit feedback, **IF:** Implicit feedback,**T:** Timestamps, **SD:** Session data, **Cx:** Contextual features, **S:** System data, **FC:** Feedback and control signals, **SG:** Social graph, **O:** Other external knowledge

Feedback and control signals. This data element captures user responses specifically related to system-generated recommendations, as well as direct inputs that influence what may be recommended. It includes exposure logs that record which items were presented as recommendations, platform-enabled behavioural responses (*e.g.*, commenting on, sharing, following, or saving recommended content), and user controls and preferences that shape the recommendation process (*e.g*, blocking items, or expressing dislikes). Explicit user surveys on recommendation satisfaction are also included in this category. These signals are distinct from general interaction data, which record all user–item engagement regardless of whether items were recommended. Feedback and control signals typically require online or instrumented environments and were only present in the MIND data set in the form of impression logs.

External knowledge sources. This data element encompasses any external data that can be integrated into the recommender system to enhance its performance or interpretability. This includes knowledge graphs (*e.g.*, DBpedia, WikiData), *linked open data* (LOD), social networks, geospatial data, cross-domain item relations and market-level data on competitive dynamics [39]. Only four of the data sets reviewed include social graph information which facilitate the modelling of social influence or trust-aware recommendation. MovieLens 32M provides direct links to external structured knowledge sources and MIND includes entity embeddings from WikiData.

This analysis highlights both the richness and the limitations of publicly available data sets for recommender systems. While all reviewed data sets provide the core elements required to model user-item interactions, the availability of additional attributes remains highly variable. The secondary data elements are rarely included, primarily due to their dependence on online system infrastructure. Common gaps include the absence of provider identifiers, limited support for session-based or contextual modelling, and a general lack of data capturing user responses to recommendations. These omissions constrain the scope of evaluations in offline settings, particularly for stakeholder-oriented values such as exposure fairness, provider visibility, or user control.

4 STAKEHOLDER REPRESENTATION IN PUBLIC DATA SETS

In MSRSs, the identification of relevant stakeholders requires careful consideration of both the application context and the available

data. In this section, we map publicly available data sets to stakeholder roles, using the proposed data taxonomy of the previous section and the stakeholder categorisation proposed at the Dagstuhl Seminar, discussed in §2.

The domain and business model of a recommender system play a critical role in determining which stakeholders are involved and which value dimensions should be prioritised. Our data set selection spans a diverse range of application domains, including media and entertainment, news and information, publishing and literature, healthcare and wellness, education and e-learning, job portals and career services, e-commerce and retail, and travel and hospitality. Within each domain, business models may differ significantly. For example, both MovieLens and Netflix pertain to movie recommendations, yet MovieLens functions as a rating platform, whereas Netflix operates as a commercial streaming service. While the former focuses primarily on capturing user preferences, the latter must also consider provider visibility and content diversity, often within a regulatory framework. Understanding the interplay between domain and business model is therefore essential to order to identify the relevant stakeholders and their associated values.

We analysed the presence of specific data elements required to support each role in order to map stakeholder roles to data sets, as defined in the taxonomy in Section 3. For each data set, we identified whether the necessary data types were available to support direct modelling or evaluation of a given stakeholder's objectives. *Consumer* roles were considered supported when user identifiers and interaction logs were available. In practice, recommender systems implicitly serve system objectives by maximising consumer satisfaction, which often aligns with business goals such as engagement, retention, or conversion. In our analysis, however, the *system* stakeholder was only acknowledged when explicit data enabling the evaluation of system-level outcomes were available, such as descriptive features together with a monetary value, or system data (*e.g.*, impression logs, KPIs). *Provider* roles required provider information such as author, artist, or seller metadata. Furthermore, *upstream* stakeholders were inferred from provider metadata or external knowledge sources, while *downstream* stakeholders were considered only when user-to-user social graphs were present. *Third-party* perspectives (e.g., advertisers, regulators) could not be represented in any of the data sets.

Table 2 contains the stakeholder roles that may be considered for evaluation in a subset of the data sets[4]—specifically, those where at least one stakeholder role beyond the consumer is directly supported by the available data. Of the data sets included, 7 contain data supporting direct evaluation of the system as a stakeholder, 11 contain provider metadata, 4 include social or trust networks that enable downstream stakeholder analysis, and only 2 support upstream stakeholders. Data sets from transaction-based domains, such as e-commerce and retail, often include price information, which supports the modelling and evaluation of system-oriented strategies such as generating profit-aware recommendations.

While Table 2 provides a useful overview of the data available to support the evaluation of different stakeholder roles, it should be interpreted as an indicator of data availability rather than a definitive guide to stakeholder importance. The table does not account for the domain or business model of each system, which play a critical role in identifying the stakeholders that have to be considered. The presence of stakeholder-related data facilitates evaluation, but the appropriateness and priority of those evaluations depend sensitively on the broader application context. In order to assess stakeholder impact meaningfully, data sets should be considered not only in terms of the available data but also in relation to the specific goals and constraints of the underlying domain.

Table 2: Stakeholder roles directly supported by the available data in data sets containing information for multiple stakeholder groups.

Data set	C	S	P	U	D
KuaiRec	×		×		×
Steam	×	×	×		
Last.FM	×		×		×
Spotify MPD	×		×		
MIND	×	×			
Adressa	×		×		
Goodreads	×		×		×
Bookcrossing	×		×	×	
Amazon2023	×	×	×	×*	
AmazonM2	×	×	×		
Epinions	×	×			×
Diginetica	×	×			
Synerise	×	×			
Yelp	×		×		
Tripadvisor	×		×		

Note
* not consistent for all item categories
C: Consumer, **S:** System, **P:** Provider, **U:** Upstream, **D:** Downstream

5 DATA REQUIREMENTS FOR VALUE REALISATION

Stakeholder values in recommender systems are not directly measurable and should be operationalised as goals that can be assessed in terms of appropriate metrics. This requires suitable data. In this section, we outline the data required to evaluate key stakeholder values, thereby providing a foundation for understanding and utilising data sets during the design of reproducible, value-aligned evaluation pipelines.

The values that we examined were drawn from the *Economic and Business* and *Societal and Human-centric* categories proposed in the Dagstuhl Seminar [8], which offers the most comprehensive list of stakeholder values currently available. Their identification is grounded in an extensive review of existing research streams, stakeholder frameworks, and application domains. All values in these two categories were examined in order to assess their feasibility for evaluation purposes.

Since recommender systems research is predominantly conducted in offline environments, evaluation is often limited to what is observable in logged interaction data or static metadata [49]. This restricts value assessment to objectives that can be quantified in

[4]An extended version of Table 2 is available at https://anonymous.4open.science/r/Multi-Stakeholder-RecSys-Datasets-6FD5.

terms of proxy metrics or structural indicators within the data set, such as rating prediction for user satisfaction, or exposure disparity as a proxy for fairness.

We systematically mapped stakeholder values to their additional data requirements beyond standard user-item interaction logs, in order to support and promote future research in multistakeholder evaluation. For each value, we identified:

(1) A measurable system-level objective aligned with the value,
(2) the primary stakeholder groups concerned,
(3) an offline proxy for evaluating the goal, where feasible, and
(4) the additional data required to support this evaluation.

These mappings are summarised in Table 3 (Appendix A) and provide a structured foundation for principled data set selection and evaluation pipeline design, thereby equipping researchers to conduct more transparent and value-aligned multistakeholder evaluations.

It is important to note that certain stakeholder values can be quantified meaningfully using only standard interaction logs, system-level infrastructure data, and model-derived information (e.g., learned item embeddings). These include *user perception and customer loyalty*, *influencing user behaviour to create demand*, *efficiency*, *novelty and product planning*, and *environmental sustainability*. For these values, proxy metrics such as accuracy scores, long-tail exposure rates, or computational resource indicators can be derived directly from offline logs and model outputs without requiring additional data. This highlights the feasibility of incorporating certain stakeholder objectives into offline evaluations, even in settings with limited data availability.

6 DESIGNING REPRODUCIBLE MULTISTAKEHOLDER EVALUATIONS

By systematically mapping data requirements to stakeholder values, we support both data-driven (bottom-up) and value-driven (top-down) evaluation design. In this section, we outline how our work may be applied to offer guidance for data set selection, proxy definition, and evaluation planning.

6.1 Guidelines for Data set Selection and Evaluation Design

In top-down scenarios, a recommender system may be designed to uphold specific stakeholder values. One may begin by selecting a value, identifying its associated stakeholders and offline proxies from our framework, and tracing backward to determine the data requirements. This supports principled data set construction, system design, or feature augmentation aimed at facilitating value-aligned evaluation.

In the bottom-up scenario, the data set is taken as the starting point, making it particularly relevant for reproducible research, where experiments are typically constrained to publicly available data sets. One should begin by identifying the present data elements according to the taxonomy in §3. Based on these elements, supported stakeholder roles may be inferred (§4). Our value-data mapping (Table 3) then enables identification of which stakeholder values might be evaluated meaningfully by using offline proxies.

In order to promote transparency and reproducibility, we recommend that researchers explicitly:

(1) Indicate which data elements are utilised,
(2) specify the stakeholder groups supported by the data set,
(3) document assumptions in value-to-proxy mappings, and
(4) justify selected evaluation metrics.

6.2 Example of Data-centric Evaluation Design

We present three representative data sets in order to demonstrate the applicability of our stakeholder–value mapping, each supporting a different configuration of stakeholder roles in Table 2. The examples selected collectively cover consumer, system, provider, upstream, and downstream stakeholders, thereby illustrating the breadth of our framework. For each data set, we describe its domain, business model, available data elements, and the stakeholder values that can be evaluated by means of offline proxies.

Goodreads operates in the publishing and literature domain as a community-driven, subscription-free platform. It provides rich item metadata (*e.g.*, genre, author), explicit feedback (ratings, reviews), implicit signals (reading behaviour), and social structures (e.g., book clubs). For consumers, it supports evaluation of *usefulness* via rating prediction accuracy and *personal growth* via content diversity in recommendations. For providers (authors), *recognition* may be measured in respect of recommendation exposure distribution across authors. For downstream stakeholders (book clubs), *community bonding* may be considered via recommendation overlap within social groups.

Steam operates in the media and entertainment domain as a transactional gaming platform and a digital distribution service. It provides detailed item metadata (*e.g.*, genre, price, release date), developer and publisher information, and explicit feedback (user reviews). For consumers, *usefulness* may be quantified in terms of ranking accuracy. For the system stakeholder, *monetary reward* may be evaluated by aligning recommendations with high-value items (using price data). For providers (developers or publishers), *fairness* may be assessed via exposure distribution.

Amazon2023 spans the e-commerce domain and provides rich item metadata, including category, price, descriptive features, and specifications, with available fields varying by product category. In the *beauty* category, for example, attributes such as brand and manufacturer facilitate partial inference of upstream stakeholders. The data set includes explicit feedback (ratings, reviews) and link structures (e.g., bought-together graphs). For consumers, *usefulness* can be assessed via rating prediction accuracy. For system stakeholders, *complementarities*, *cross-selling*, and *monetary reward* may be evaluated in terms of co-purchase links and price-based item value. *Recognition and fairness* for providers (stores), and *reputation* for upstream actors (brands or manufacturers) may be examined by considering exposure distribution in recommendations.

7 LIMITATIONS AND FUTURE WORK

While we presented a structured approach in this paper towards MSRS evaluation evaluation grounded in concrete data requirements, several limitations are acknowledged.

First, our value–data mappings rely on offline approximations which may not fully capture the nuances of real-world system dynamics. The offline proxies proposed are therefore best viewed as coarse approximators rather than definitive measures. Moreover, the current value–data mapping is not exhaustive: While it covers a broad range of stakeholder values, additional goals and their offline proxies may be defined for the existing values. New values may emerge as the field evolves or as novel application domains are introduced. Researchers are encouraged to follow the guidelines outlined in Section 6.1, in order to incorporate such extensions, which would support the transparent reporting and integration of newly defined values, goals, and offline proxies into multi-stakeholder evaluation pipelines.

Moreover, our analysis was limited to 30 publicly available data sets. Although selected for domain diversity and relevance, this sample does not represent the full spectrum of recommender system data sets.

While our taxonomy and mappings aim to be generalisable, evaluation design remains context-dependent. The salience of values and the suitability of proxies can vary across application domains, cultural settings, and business models. Our framework should thus be viewed as a guide rather than as a prescriptive template.

Our work does not focus on the development or benchmarking of evaluation metrics. For a detailed discussion on metric construction and validation in multistakeholder settings, we refer the reader to the contributions of Burke *et al.* in the Dagstuhl Seminar report [8].

Future work may apply our systematic methodology to extend existing evaluation frameworks, such as RecBole [50] and Elliot [6], to include multiple stakeholders. We also highlight the need for new data sets that include the required data elements for supporting robust MSRS evaluation. In the absence of such data, synthetic environments offer a promising direction for simulating stakeholder roles and testing evaluation strategies under controlled conditions.

8 CONCLUSION

Our work builds directly on the conceptual foundations proposed by the Dagstuhl Seminar [8], extending them with concrete, data-driven tools aimed at supporting MSRS evaluation. By mapping stakeholder values to offline proxies and minimum data requirements, we provide a flexible framework that supports both bottom-up and top-down evaluation design. The methodology enables researchers and practitioners to integrate stakeholder-awareness into existing toolkits, guide data set selection, and document evaluation assumptions more transparently. Whether embedded into benchmarking libraries or used as a structured reporting checklist, our proposed approach supports more principled and reproducible evaluation of MSRSs.

REFERENCES

[1] 2015. Retailrocket eCommerce Dataset. https://www.kaggle.com/datasets/retailrocket/ecommerce-dataset. Accessed 2025.

[2] 2016. CIKM Cup 2016 Track 2: Personalized E-Commerce Search Challenge. https://competitions.codalab.org/competitions/11161. Accessed 2025.

[3] 2024. TripAdvisor Hotel Reviews Dataset. https://www.kaggle.com/datasets/joebeachcapital/hotel-reviews. Accessed 2025.

[4] Himan Abdollahpouri, Gediminas Adomavicius, Robin Burke, Ido Guy, Dietmar Jannach, Toshihiro Kamishima, Jan Krasnodebski, and Luiz Pizzato. 2020. Multistakeholder recommendation: Survey and research directions. *User Model. User-adapt Interact.* 30 (2020), 127–158.

[5] Himan Abdollahpouri and Robin Burke. 2019. Multi-stakeholder Recommendation and its Connection to Multi-sided Fairness. In *Proceedings of the Workshop on Recommendation in Multistakeholder Environments (RMSE) co-located with the 13th ACM Conference on Recommender Systems (RecSys '19)*. Copenhagen, Denmark.

[6] Vito Walter Anelli, Alejandro Bellogin, Antonio Ferrara, Daniele Malitesta, Felice Antonio Merra, Claudio Pomo, Francesco Maria Donini, and Tommaso Di Noia. 2021. Elliot: A Comprehensive and Rigorous Framework for Reproducible Recommender Systems Evaluation. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval* (Virtual Event, Canada) *(SIGIR '21)*. Association for Computing Machinery, New York, NY, USA, 2405–2414. https://doi.org/10.1145/3404835.3463245

[7] Nabiha Asghar. 2016. Yelp Dataset Challenge: Review Rating Prediction. arXiv:1605.05362 https://arxiv.org/abs/1605.05362

[8] Christine Bauer, Alan Said, and Eva Zangerle. 2024. *Evaluation Perspectives of Recommender Systems: Driving Research and Education (Dagstuhl Seminar 24211)*. Dagstuhl Reports 5. Schloss Dagstuhl – Leibniz-Zentrum für Informatik. 58–172 pages.

[9] Joeran Beel, Corinna Breitinger, Stefan Langer, Andreas Lommatzsch, and Bela Gipp. 2016. Towards reproducibility in recommender-systems research. *User Modeling and User-Adapted Interaction* 26 (2016), 69–101. https://doi.org/10.1007/s11257-016-9174-x

[10] Joeran Beel, Alan Said, Tobias Vente, and Lukas Wegmeth. 2024. Green Recommender Systems: A Call for Attention. *ACM SIGIR Forum* 58, 2 (2024). https://isg.beel.org/pubs/2024_Green_Recommender_Systems-A_Call_for_Attention.pdf

[11] Robin Burke and Himan Abdollahpouri. 2017. Patterns of Multistakeholder Recommendation. arXiv:1707.09258 https://arxiv.org/abs/1707.09258

[12] Robin Burke, Gediminas Adomavicius, Toine Bogers, Tommaso Di Noia, Dominik Kowald, Julia Neidhardt, Özlem Özgöbek, Maria Soledad Pera, Nava Tintarev, and Jürgen Ziegler. 2025. De-centering the (Traditional) User: Multistakeholder Evaluation of Recommender Systems. arXiv:2501.05170 https://arxiv.org/abs/2501.05170

[13] Iván Cantador, Peter Brusilovsky, and Tsvi Kuflik. 2011. 2nd Workshop on Information Heterogeneity and Fusion in Recommender Systems (HetRec 2011). In *Proceedings of the 5th ACM conference on Recommender systems* (Chicago, IL, USA) *(RecSys 2011)*. ACM, New York, NY, USA.

[14] Ching-Wei Chen, Paul Lamere, Markus Schedl, and Hamed Zamani. 2018. Recsys challenge 2018: automatic music playlist continuation. In *Proceedings of the 12th ACM Conference on Recommender Systems* (Vancouver, British Columbia, Canada) *(RecSys '18)*. Association for Computing Machinery, New York, NY, USA, 527–528. https://doi.org/10.1145/3240323.3240342

[15] Youngduck Choi, Youngnam Lee, Dongmin Shin, Junghyun Cho, Seoyon Park, Seewoo Lee, Jineon Baek, Chan Bae, Byungsoo Kim, and Jaewe Heo. 2020. EdNet: A Large-Scale Hierarchical Dataset in Education. In *Artificial Intelligence in Education*, Ig Ibert Bittencourt, Mutlu Cukurova, Kasia Muldner, Rose Luckin, and Eva Millán (Eds.). Springer International Publishing, Cham, 69–73.

[16] José A. Concha-Carrasco, Miguel A. Vega-Rodríguez, and Carlos J. Pérez. 2023. A multi-objective artificial bee colony approach for profit-aware recommender systems. *Information Sciences* 625 (2023), 476–488. https://doi.org/10.1016/j.ins.2023.01.050

[17] Paolo Cremonesi and Dietmar Jannach. 2021. Progress in recommender systems research: Crisis? What crisis? *AI Magazine* 42, 3 (2021), 43–54. https://doi.org/10.1609/aimag.v42i3.18145

[18] Alvise De Biasio, Dietmar Jannach, and Nicolò Navarin. 2024. Model-based approaches to profit-aware recommendation. *Expert Syst. Appl.* 249, Article 123642 (Sept. 2024).

[19] Alvise De Biasio, Andrea Montagna, Fabio Aiolli, and Nicolò Navarin. 2023. A systematic review of value-aware recommender systems. *Expert Syst. Appl.* 226, Article 120131 (Sept. 2023).

[20] Alvise De Biasio, Nicolò Navarin, and Dietmar Jannach. 2024. Economic recommender systems – a systematic review. *Electron. Commer. Res. Appl.* 63, Article 101352 (Jan. 2024).

[21] Yashar Deldjoo, Dietmar Jannach, Alejandro Bellogin, Alessandro Difonzo, and Dario Zanzonelli. 2023. Fairness in recommender systems: research landscape and future directions. *User Modeling and User-Adapted Interaction* 34, 1 (April 2023), 59–108. https://doi.org/10.1007/s11257-023-09364-z

[22] Dieter Fensel, Katia Sycara, and John Mylopoulos (Eds.). 2003. *Trust Management for the Semantic Web*. Springer Berlin Heidelberg, Berlin, Heidelberg. https://doi.org/10.1007/978-3-540-39718-2_23

[23] Chongming Gao, Shijun Li, Wenqiang Lei, Jiawei Chen, Biao Li, Peng Jiang, Xiangnan He, Jiaxin Mao, and Tat-Seng Chua. 2022. KuaiRec: A Fully-observed Dataset and Insights for Evaluating Recommender Systems. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management* (Atlanta, GA, USA) *(CIKM '22)*. Association for Computing Machinery, New York, NY, USA, 540–550. https://doi.org/10.1145/3511808.3557220

[24] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. 2021. Datasheets for datasets. *Commun. ACM* 64, 12 (Nov. 2021), 86–92. https://doi.org/10.1145/3458723

[25] Jon Atle Gulla, Lemei Zhang, Peng Liu, Özlem Özgöbek, and Xiaomeng Su. 2017. The Adressa dataset for news recommendation. In *Proceedings of the International Conference on Web Intelligence* (Leipzig, Germany) *(WI '17)*. Association for Computing Machinery, New York, NY, USA, 1042–1048. https://doi.org/10.1145/3106426.3109436

[26] F. Maxwell Harper and Joseph A. Konstan. 2015. The MovieLens Datasets: History and Context. *ACM Trans. Interact. Intell. Syst.* 5, 4, Article 19 (Dec. 2015), 19 pages. https://doi.org/10.1145/2827872

[27] Yupeng Hou, Jiacheng Li, Zhankui He, An Yan, Xiusi Chen, and Julian McAuley. 2024. Bridging Language and Items for Retrieval and Recommendation. *arXiv preprint arXiv:2403.03952* (2024).

[28] Dietmar Jannach and Himan Abdollahpouri. 2023. A survey on multi-objective recommender systems. *Front. Big Data* 6, Article 1157899 (2023).

[29] Dietmar Jannach and Michael Jugovac. 2019. Measuring the Business Value of Recommender Systems. *ACM Trans. Manage. Inf. Syst.* 10, 4, Article 16 (2019), 23 pages. https://doi.org/10.1145/3370082

[30] Dietmar Jannach and Markus Zanker. 2022. Value and Impact of Recommender Systems. In *Recommender Systems Handbook*. Springer Nature.

[31] Wei Jin, Haitao Mao, Zheng Li, Haoming Jiang, Chen Luo, Hongzhi Wen, Haoyu Han, Hanqing Lu, Zhengyang Wang, Ruirui Li, Zhen Li, Monica Cheng, Rahul Goutam, Haiyang Zhang, Karthik Subbian, Suhang Wang, Yizhou Sun, Jiliang Tang, Bing Yin, and Xianfeng Tang. 2023. Amazon-M2: a multilingual multi-locale shopping session dataset for recommendation and text generation. In *Proceedings of the 37th International Conference on Neural Information Processing Systems* (New Orleans, LA, USA) *(NIPS '23)*. Curran Associates Inc., Red Hook, NY, USA, Article 351, 21 pages.

[32] Wang-Cheng Kang and Julian McAuley. 2018. Self-Attentive Sequential Recommendation. arXiv:1808.09781 https://arxiv.org/abs/1808.09781

[33] Johannes Kruse, Kasper Lindskow, Saikishore Kalloori, Marco Polignano, Claudio Pomo, Abhishek Srivastava, Anshuk Uppal, Michael Riis Andersen, and Jes Frellsen. 2024. EB-NeRD a large-scale dataset for news recommendation. In *Proceedings of the Recommender Systems Challenge 2024* (Bari, Italy) *(RecSysChallenge '24)*. Association for Computing Machinery, New York, NY, USA, 1–11. https://doi.org/10.1145/3687151.3687152

[34] Elio Masciari, Areeba Umair, and Muhammad Habib Ullah. 2024. A Systematic Literature Review on AI-Based Recommendation Systems and Their Ethical Considerations. *IEEE Access* 12 (2024), 121223–121241. https://doi.org/10.1109/ACCESS.2024.3451054

[35] Silvia Milano, Mariarosaria Taddeo, and Luciano Floridi. 2020. Recommender systems and their ethical challenges. *AI & SOCIETY* 35, 4 (2020), 957–967. https://doi.org/10.1007/s00146-020-00950-y

[36] Netflix Prize. 2009. The Netflix Prize. http://www.netflixprize.com/. Retrieved in 2025 from https://www.kaggle.com/datasets/netflix-inc/netflix-prize-data.

[37] Jianmo Ni, Larry Muhlstein, and Julian McAuley. 2019. Modeling Heart Rate and Activity Data for Personalized Fitness Recommendation. In *The World Wide Web Conference* (San Francisco, CA, USA) *(WWW '19)*. Association for Computing Machinery, New York, NY, USA, 1343–1353. https://doi.org/10.1145/3308558.3313643

[38] Sergio Oramas, Vito Claudio Ostuni, Tommaso Di Noia, Xavier Serra, and Eugenio Di Sciascio. 2016. Sound and Music Recommendation with Knowledge Graphs. *ACM Trans. Intell. Syst. Technol.* 8, 2, Article 21 (Oct. 2016). https://doi.org/10.1145/2926718

[39] Francesco Ricci, Lior Rokach, and Bracha Shapira (Eds.). 2022. *Recommender Systems Handbook*. Springer Nature.

[40] Deepjyoti Roy and Mala Dutta. 2022. A systematic review and research perspective on recommender systems. *Journal of Big Data* 9, Article 59 (2022), 36 pages. https://doi.org/10.1186/s40537-022-00592-5

[41] Thiago Silveira, Min Zhang, Xiao Lin, Yiqun Liu, and Shaoping Ma. 2019. How good your recommender system is? A survey on evaluations in recommendation. *International Journal of Machine Learning and Cybernetics* 10, 5 (2019), 813–831. https://doi.org/10.1007/s13042-017-0762-9

[42] somnambwl. 2020. Book-Crossing Dataset. https://www.kaggle.com/datasets/somnambwl/bookcrossing-dataset. Accessed 2025.

[43] Özge Sürer, Robin Burke, and Edward C. Malthouse. 2018. Multistakeholder recommendation with provider constraints. In *Proceedings of the 12th ACM Conference on Recommender Systems* (Vancouver, British Columbia, Canada) *(RecSys '18)*. Association for Computing Machinery, New York, NY, USA, 54–62. https://doi.org/10.1145/3240323.3240350

[44] Synerise. 2025. RecSys Challenge 2025: Universal Behavioral Profiles Dataset. https://recsys.synerise.com. Accessed 2025.

[45] Mengting Wan and Julian McAuley. 2018. Item recommendation on monotonic behavior chains. In *Proceedings of the 12th ACM Conference on Recommender Systems* (Vancouver, British Columbia, Canada) *(RecSys '18)*. Association for Computing Machinery, New York, NY, USA, 86–94. https://doi.org/10.1145/3240323.3240369

[46] Mengting Wan, Rishabh Misra, Ndapa Nakashole, and Julian J. McAuley. 2019. Fine-Grained Spoiler Detection from Large-Scale Review Corpora. In *Proceedings of the 57th Conference of the Association for Computational Linguistics, ACL 2019, Florence, Italy, July 28- August 2, 2019, Volume 1: Long Papers*, Anna Korhonen, David R. Traum, and Lluís Màrquez (Eds.). Association for Computational Linguistics, 2605–2610. https://doi.org/10.18653/V1/P19-1248

[47] Yifan Wang, Weizhi Ma, Min Zhang, Yiqun Liu, and Shaoping Ma. 2023. A Survey on the Fairness of Recommender Systems. *ACM Trans. Inf. Syst.* 41, 3, Article 52 (Feb. 2023), 43 pages. https://doi.org/10.1145/3547333

[48] Fangzhao Wu, Ying Qiao, Jiun-Hung Chen, Chuhan Wu, Tao Qi, Jianxun Lian, Danyang Liu, Xing Xie, Jianfeng Gao, Winnie Wu, and Ming Zhou. 2020. MIND: A Large-scale Dataset for News Recommendation. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel Tetreault (Eds.). Association for Computational Linguistics, Online, 3597–3606. https://doi.org/10.18653/v1/2020.acl-main.331

[49] Eva Zangerle and Christine Bauer. 2022. Evaluating Recommender Systems: Survey and Framework. *ACM Comput. Surv.* 55, 8, Article 170 (2022), 38 pages. https://doi.org/10.1145/3556536

[50] Wayne Xin Zhao, Shanlei Mu, Yupeng Hou, Zihan Lin, Yushuo Chen, Xingyu Pan, Kaiyuan Li, Yujie Lu, Hui Wang, Changxin Tian, Yingqian Min, Zhichao Feng, Xinyan Fan, Xu Chen, Pengfei Wang, Wendi Ji, Yaliang Li, Xiaoling Wang, and Ji-Rong Wen. 2021. RecBole: Towards a unified, comprehensive and efficient framework for recommendation algorithms. In *Proceedings of the 30th ACM International Conference on Information & Knowledge Management* (Virtual Event, Queensland) *(CIKM '21)*. ACM Press, New York, (NY), 4653–4664. https://doi.org/10.1145/3459637.3482016

[51] Yong Zheng. 2019. Multi-stakeholder recommendations: case studies, methods and challenges. In *Proceedings of the 13th ACM Conference on Recommender Systems* (Copenhagen, Denmark) *(RecSys '19)*. Association for Computing Machinery, New York, NY, USA, 578–579. https://doi.org/10.1145/3298689.3346951

[52] Yu Zheng, Chen Gao, Xiangnan He, Yong Li, and Depeng Jin. 2020. Price-aware Recommendation with Graph Convolutional Networks. In *IEEE 36th International Conference on Data Engineering (ICDE)*. 133–144. https://doi.org/10.1109/ICDE48307.2020.00019

A VALUE-DATA MAPPINGS

Table 3: Offline evaluation proxies and additional data requirements for stakeholder values

Value	Stakeholders	Example Goal	Offline Evaluation Proxy	Min Data Requirements
Economic and Business Values				
Monetary reward	System, Providers	Maximise economic returns by promoting high-value and demand-driven items.	Assess whether high-margin or high-priced items are recommended.	Items: Descriptive features (e.g., price)
User Perception and Customer Loyalty	System	Increase user retention and loyalty by improving perceived relevance and satisfaction.	Use accuracy metrics (e.g., RMSE, Precision@k) to approximate user satisfaction.	–
Growth and Market Development	System	Support platform growth by acquiring new users or markets with recommendations.	*No reliable offline proxy—requires real-time tracking of user acquisition.*	Feedback and control signals
Influencing User Behaviour to Create Demand	System	Encourage preference shifts by exposing users to less familiar or long-tail content.	Approximate with long-tail exposure or inverse item popularity of recommendations.	–
Efficiency	System	Maximise engagement or exposure while using minimal computational or operational resources.	Assess trade-off between resource use and recommendation quality.	System data: Algorithm metadata (e.g., runtime, latency)
Novelty and Product Planning	System, Consumers, Providers	Identify which novel items resonate with users to inform product or content strategy.	Measure exposure to items that differ from a user's historical preferences.	Items: Descriptive features
Complementarities	System	Promote synergistic consumption patterns by recommending complementary items.	Estimate co-consumption strength using lift or confidence between item pairs.	Interactions: Timestamps or Session data; Items: Descriptive features (e.g., category)
Lock-in and Churn Prevention	System	Prevent churn by maintaining long-term user engagement.	*No reliable offline proxy—requires long-term feedback or retention tracking.*	Feedback and control signals
Brand Equity	System, Providers	Strengthen brand value by reinforcing positive user experiences and trust.	*No reliable offline proxy—requires perception data or long-term brand impact tracking.*	Feedback and control signals
Cross and Upselling	System, Providers	Increase basket size or transaction value through strategic cross- or up-sell recommendations.	Evaluate whether co-consumed or higher-value items are recommended in the same or subsequent session.	Interactions: Timestamps, Session data; Items: Descriptive features (e.g., price, category)
Personalised Marketing Strategies	System	Support personalised outreach and retention strategies based on user preferences and behaviour.	*No reliable offline proxy—requires integration with marketing outcomes and user consent mechanisms.*	Feedback and control signals (e.g., user preferences, opt-ins)
Societal and Human-centric Values				
Usefulness and Enjoyment	Consumers	Increase user satisfaction by recommending relevant and enjoyable content.	Approximate usefulness through rating prediction or ranking accuracy as a measure of alignment with past preferences.	–
Control and Privacy	Consumers, Third-party	Empower users to control preferences, data sharing, and personalisation settings.	*No reliable offline proxy—requires data on user controls or privacy configurations.*	Feedback and control signals
Well-being	Consumers, Third-party	Ensure recommendations contribute positively to user well-being and avoid harmful content.	*No reliable offline proxy—requires subjective feedback or content impact monitoring.*	Feedback and control signals; Items: Descriptive features (e.g., content type)
Connection, Community & Social Bonding	Consumers, Providers	Strengthen user or creator communities through socially aligned recommendations.	Evaluate recommendation overlap or co-engagement within social groups.	External knowledge: Social graphs
Recognition, Reputation and Acknowledgment	Providers, Upstream	Promote creator visibility and recognition within recommendation outputs.	Assess exposure fairness and distribution of recommendations across content creators.	Items: Provider information
Personal Growth and Development	Consumers	Support user development by broadening content exposure and interests.	Evaluate diversity and novelty in recommendations.	Items: Descriptive features (e.g., category)
Fairness	Consumers, Providers, Third-party	Ensure equal opportunity and non-discrimination in recommendation delivery.	Audit disparities in exposure or performance across stakeholder groups.	Users: Demographics Items: Provider information
Accessibility & Inclusion	Consumers, Third-party	Promote inclusive access and equitable content delivery for diverse user groups.	Evaluate recommendation exposure and performance across user demographics and accessible content features.	Users: Demographics (e.g., disability status); Items: Descriptive features (e.g., accessibility tags)
Freedom of Expression	Providers, Upstream	Ensure that diverse creators can reach audiences and express ideas freely.	Assess exposure of niche or non-mainstream content in recommendations.	Items: Descriptive features (e.g., category); Provider information
Transparency and Trustworthiness	All	Build stakeholder trust by making recommendations explainable and interpretable.	Evaluate whether recommendations can be justified using available metadata.	Items: Descriptive features (applicable when explainable methods are employed)
Social Benefit	Third-party, Consumers	Ensure recommendations contribute to positive societal outcomes such as education, health, or cultural cohesion.	*No reliable offline proxy—requires real-world outcome tracking or longitudinal user impact studies.*	Feedback and control signals; External knowledge sources (e.g., public health or cultural indicators)
Tradition and History	Consumers, Providers, Third-party	Promote access to culturally significant, historical, or traditional content.	Evaluate representation of heritage-aligned content.	Items: Descriptive features (e.g., release date, region, cultural tag)
Environmental Sustainability	All	Reduce carbon footprint and energy usage of system.	Evaluate computational efficiency of recommendation models.	System data: Algorithm metadata (e.g., runtime, FLOPs, energy consumption)
Safety	Consumers, Third-party	Reduce user exposure to harmful, unsafe, or inappropriate recommendations.	Detect and audit flagged or sensitive content in recommendations.	Users: Demographics (e.g., age); Items: Descriptive features (e.g., content flags, descriptions)